1A, 30V - 60V Schottky Barrier Surface Mount Rectifier

FEATURES

- Very low profile - typical height of 0.68mm
- Low power loss, high efficiency
- Ideal for automated placement
- Moisture sensitivity level: level 1, per J-STD-020
- RoHS Compliant
- Halogen-free according to IEC 61249-2-21

APPLICATIONS

- Switching mode power supply (SMPS)
- Adapters
- DC to DC converter

MECHANICAL DATA

- Case: Micro SMA
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: Indicated by cathode band
- Weight: 0.006g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	1	A
V_{RRM}	30 - 60	V
I_{FSM}	25	A
$T_{J\ MAX}$	150	°C
Package	Micro SMA	
Configuration	Single die	

Pb-Free RoHS COMPLIANT HALOGEN FREE

Micro SMA

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)					
PARAMETER	**SYMBOL**	**SS13M**	**SS14M**	**SS16M**	**UNIT**
Marking code on the device		A	B	C	
Repetitive peak reverse voltage	V_{RRM}	30	40	60	V
Reverse voltage, total rms value	$V_{R(RMS)}$	21	28	42	V
Forward current	I_F	1			A
Surge peak forward current, 8.3ms single half sine wave superimposed on rated load	I_{FSM}	25			A
Junction temperature	T_J	-55 to +150			°C
Storage temperature	T_{STG}	-55 to +150			°C

THERMAL PERFORMANCE			
PARAMETER	**SYMBOL**	**TYP**	**UNIT**
Junction-to-lead thermal resistance	$R_{\Theta JL}$	30	°C/W
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	125	°C/W
Junction-to-case thermal resistance	$R_{\Theta JC}$	40	°C/W

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)						
PARAMETER		**CONDITIONS**	**SYMBOL**	**TYP**	**MAX**	**UNIT**
Forward voltage[1]	SS13M SS14M	I_F = 0.5A,T_J = 25°C	V_F	0.45	-	V
		I_F = 1.0A,T_J = 25°C		0.52	0.55	V
		I_F = 0.5A,T_J = 125°C		0.35	-	V
		I_F = 1.0A,T_J = 125°C		0.46	0.50	V
	SS16M	I_F = 0.5A,T_J = 25°C	V_F	0.51	-	V
		I_F = 1.0A,T_J = 25°C		0.64	0.68	V
		I_F = 0.5A,T_J = 125°C		0.46	-	V
		I_F = 1.0A,T_J = 125°C		0.57	0.60	V
Reverse current @ rated V_R[2]		T_J = 25°C	I_R	-	50	μA
		T_J = 125°C		-	10	mA
Junction capacitance	SS13M SS14M	1MHz, V_R = 4.0V	C_J	50	-	pF
	SS16M			40	-	pF

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION		
ORDERING CODE[1]	**PACKAGE**	**PACKING**
SS1xM	Micro SMA	12,000 / Tape & Reel

Notes:

1. "x" defines voltage from 30V(SS13M) to 60V(SS16M)

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

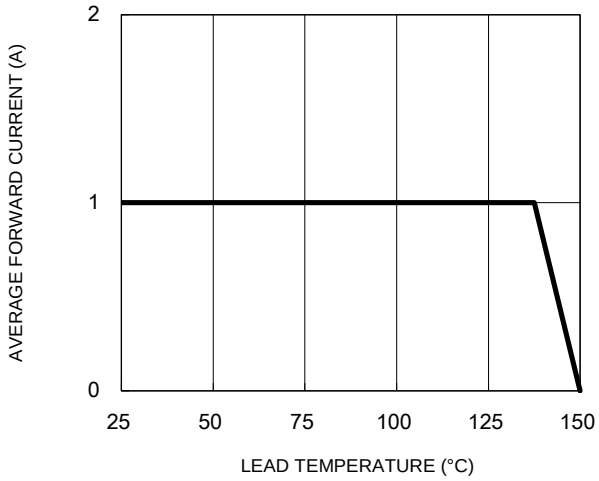


Fig.2 Typical Junction Capacitance

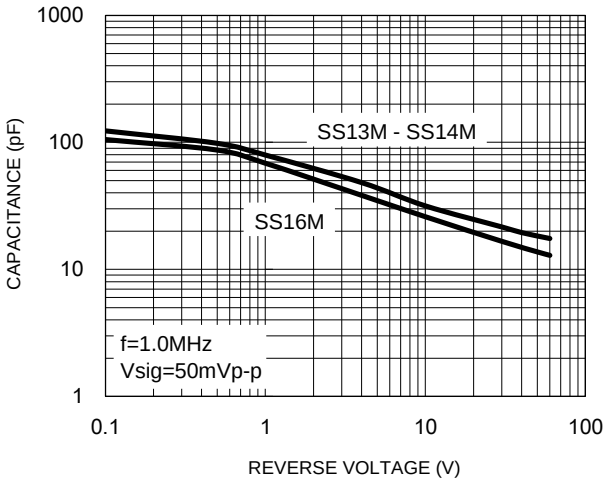


Fig.3 Typical Reverse Characteristics

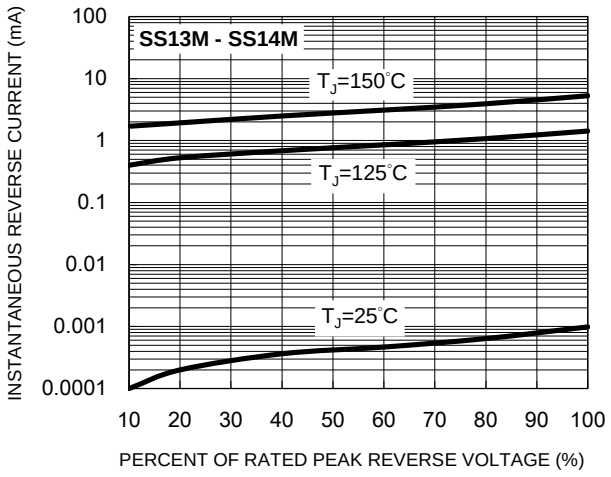


Fig.4 Typical Forward Characteristics

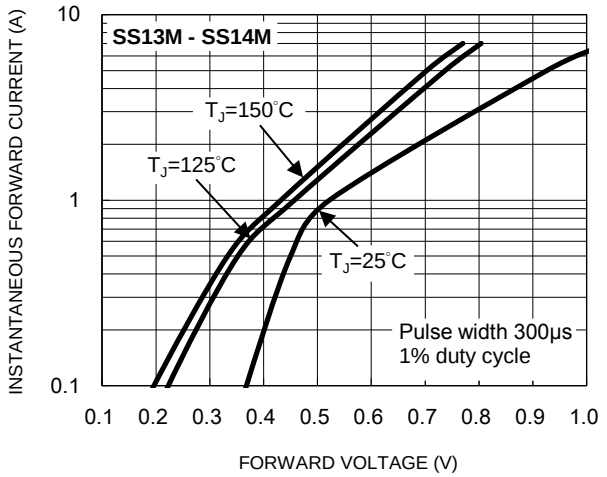


Fig.5 Typical Reverse Characteristics

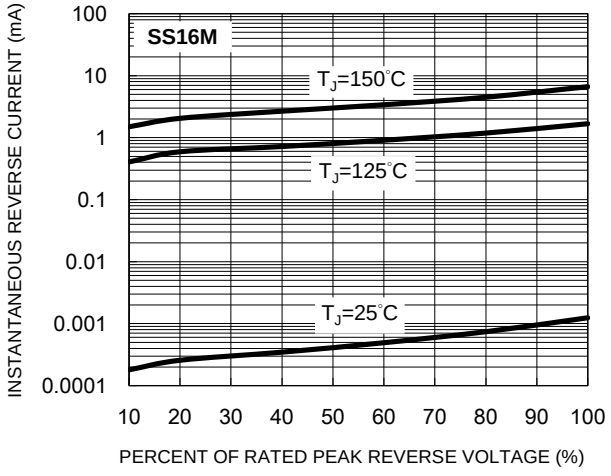
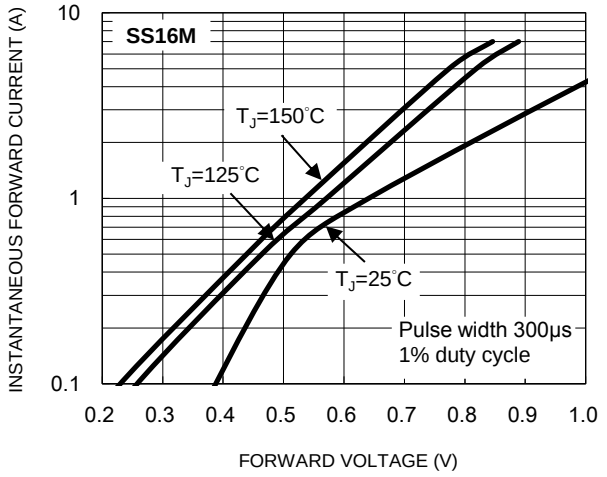


Fig.6 Typical Forward Characteristics



CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.7 Maximum Non-Repetitive Forward Surge Current

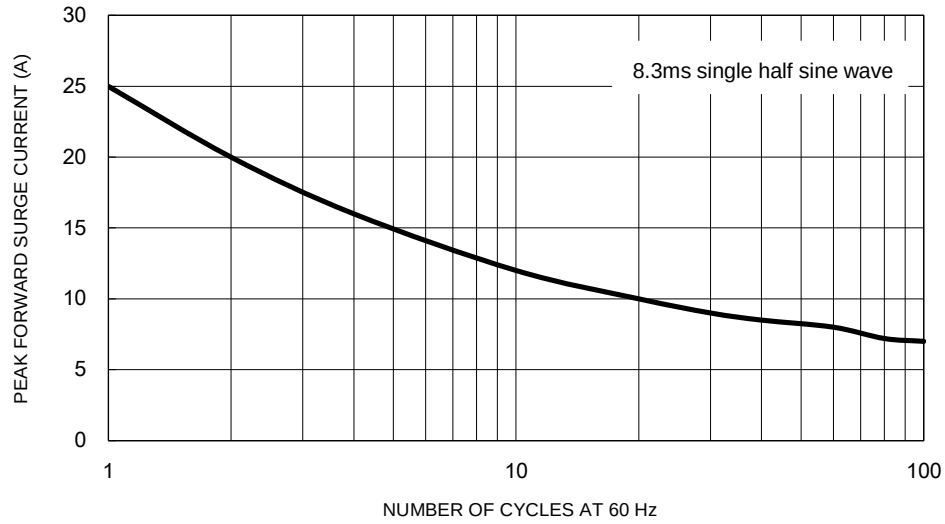
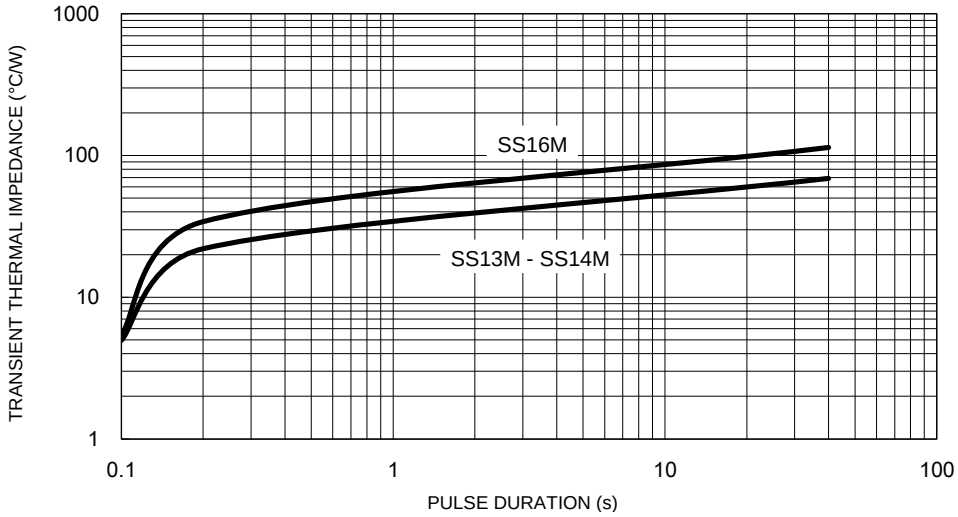
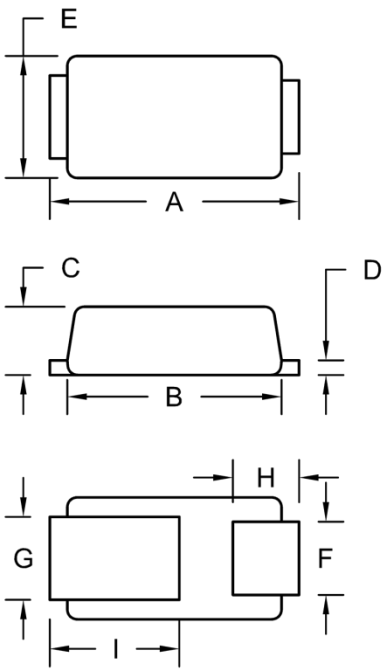


Fig.8 Typical Transient Thermal Impedance



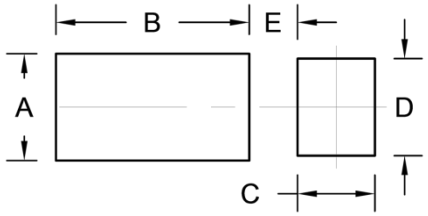
PACKAGE OUTLINE DIMENSIONS

Micro SMA



DIM.	Unit (mm)		Unit (inch)	
	Min.	Max.	Min.	Max.
A	2.30	2.70	0.091	0.106
B	2.10	2.30	0.083	0.091
C	0.63	0.73	0.025	0.029
D	0.10	0.20	0.004	0.008
E	1.15	1.35	0.045	0.053
F	0.65	0.85	0.026	0.034
G	0.75	0.95	0.030	0.037
H	0.55	0.75	0.022	0.030
I	1.10	1.50	0.043	0.059

SUGGESTED PAD LAYOUT



Symbol	Unit (mm)	Unit (inch)
A	1.10	0.043
B	2.00	0.079
C	0.80	0.031
D	1.00	0.039
E	0.50	0.020

MARKING DIAGRAM

P/N = Marking Code
YW = Data Code

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